Ashraf Hany

Education

BSc in Computer Science (Dual Degree Program) 2021 – 2025
MSA University, Egypt & University of Greenwich, UK **CGPA: 3.81/4.0**

- Graduated with **First Class Honors** from both institutions

Publications

Hany, Ashraf; Ramadan, Enjy; Akl, Amr; Atia, Ayman. *The Effect of Using Tangible User Interfaces Compared to Traditional Learning for Teaching Programming in Higher Education: An Experimental Study*. In: *2023 Intelligent Methods, Systems, and Applications (IMSA)*, 2023, pp. 514–519. DOI: `10.1109/IMSA58542.2023.10217780`

Hany, Ashraf; Gomma, W. H. *Comprehensive Benchmark: Comparing Classical, Recurrent, and Transformer-Based Models for Sentiment Analysis Across Text Length Strata*. In: *2025 Intelligent Methods, Systems, and Applications (IMSA)*, Giza, Egypt, 2025, pp. 487–494. DOI: `10.1109/IMSA65733.2025.11166950`

Ramadan, Enjy; et al. *Integrating Interactive 3D Applications for Enhanced Learning in Crown Preparations: A Comparative Experimental Study*. In: *2025 International Conference on Activity and Behavior Computing (ABC)*, Al Ain, UAE, 2025, pp. 1–10. DOI: `10.1109/ABC64332.2025.11118293`

Hany, Ashraf; Mohammed, A. *Deep Learning for Microscopic Image Segmentation in Multiple Myeloma: A Comparative Study*. In: *2025 Intelligent Methods, Systems, and Applications (IMSA)*, Giza, Egypt, 2025, pp. 256–262. DOI: `10.1109/IMSA65733.2025.11167795`

Experience

AI & Data Engineer | **Techlabs London** February 2026 – May 2026

- Designed and deployed AI-powered solutions including LLM-based conversational agents, RAG pipelines, and predictive analytics workflows using Python, LangChain, and modern ML frameworks to support intelligent automation across enterprise platforms.
- Developed scalable machine learning and computer vision pipelines for data extraction, classification, and operational insights, integrating models into production environments using Docker, REST APIs, and cloud-based infrastructure.
- Collaborated with cross-functional engineering and product teams to translate business requirements into production-ready AI features, conducting model evaluation, experimentation, and optimization to improve performance, usability, and business impact.

Teaching Assistant | **MSA University** August 2025 – Present

- Instructed **Design & Analysis of Algorithms** (5 groups, 150 students), **Embedded Systems** (2 groups, 60 students), and **Calculus 2** (2 groups, 60 students).
- Designed and delivered lab sessions, assignments, and problem-solving tutorials reinforcing theoretical concepts across all courses.
- Provided individual academic guidance to students working through programming logic and hardware implementation challenges.

***Data Analyst* - Top Rated Freelancer** | **Upwork** August 2024 – May 2026

- Designed and developed educational and scientific animations using Manim for clear visual communication of complex topics.
- Conducted AI research including implementation and evaluation of machine learning and quantum machine learning models.
- Collected, cleaned, and preprocessed data from diverse sources to build high-quality datasets.

Research Intern | **Nile University** August – September 2024

- Researched quantum machine learning (QML) models in noisy environments to improve fault tolerance.
- Analyzed and documented techniques for optimizing QML model performance under realistic noise conditions.

Quantum Machine Learning Researcher | **MSA University** February 2024 – February 2025

- Developed QML classifier models using PennyLane, benchmarked across multiple datasets as part of a university research project.

Junior Teaching Assistant (Volunteer) | **MSA University** Fall 2023

- Supported instruction for four courses: `MTH103` Discrete Math, `CS213` Multimedia Programming, `CS232` Algorithms & Data Structures, `CS102` Fundamentals of Computing II.

Cybersecurity Intern | **National Bank of Egypt** July – August 2023

- Developed a Python script for domain variant detection using symbol-oriented permutation.
- Shadowed senior professionals across Web & Network Penetration Testing, Threat Intelligence, VA, Incident Handling, and Phishing analysis.

Leadership

MSA Competitive Programming Community (CPC) — Founder & Leader ***[Link]*** ***Fall 2023 – Present***

- Founded MSA's first dedicated competitive programming community from the ground up, growing it into a recognized campus organization.
- Organized and led university participation in ECPC 2023, 2024, and 2025 with teams of 21, 28, and 28 respectively.
- Designed training programs, organized practice contests, and mentored students in competitive problem-solving strategies.

Achievements

Third Place — Engineering Track, 20th NUGRF ***[Link]*** ***Spring 2025***

- Awarded third place in the Engineering Track of Egypt's most competitive national undergraduate research fair (NU UGRF, Nile University) for designing and building a fully custom ESP32-S3 handheld game console from scratch.

Second Place — NU Quantum-AI Hackathon ***[Link]*** ***Summer 2024***

- Placed second in Egypt's first Quantum-AI Hackathon (Nile University). Preprocessed a 1389-feature Raman shift spectroscopy dataset from a Nature paper for encoding into a Variational Quantum Classifier (VQC), then built and optimized the VQC using Qiskit's RealAmplitudes ansatz, ZZFeatureMap, and COBYLA optimizer.

First Place — Best Undergraduate Project, MSA DeepMinds Event ***Spring 2023***

- Recognised as Sophomore Best Undergraduate Project for publishing an IEEE conference paper during the 2nd year of undergraduate studies.

Projects

Deep Learning-Based Image Segmentation for Multiple Myeloma | ***PyTorch, U-Net, SAM, MedSAM*** 2024 – 2025

- Conducted in collaboration with the **National Cancer Institute, Cairo University**; applied deep learning segmentation models (U-Net, SAM baseline, fine-tuned SAM, and MedSAM) to microscopic bone marrow images from the SegPC-2021 dataset to quantify plasma cell burden for treatment response evaluation.
- Proposed and evaluated a **Combined Masks** preprocessing strategy that merges cytoplasm and nucleus annotations, improving segmentation consistency across all tested architectures.
- Published findings as a first-author IEEE conference paper; results demonstrated fine-tuned SAM achieving competitive Dice scores while MedSAM provided the strongest domain-specific generalization.

ESP32-S3 Gameboy ***[Link]*** | ***C, Retro-Go, Embedded Systems, PCB Design*** Spring 2025

- Built a fully custom handheld gaming console on the ESP32-S3: 3.2-inch display, custom button PCB, MAX9837 audio DAC, and ergonomic 3D-printed enclosure.
- Ported and optimized Retro-Go firmware to emulate NES, Game Boy, and Sega platforms at full speed. Shortlisted at NUGRF 20th Edition.

IMU-Based Motion Light Gun ***[Link]*** | ***Arduino, MPU6050, Embedded Systems*** Summer 2025

- Designed a motion-controlled light gun using an Arduino Pro Mini and MPU6050 IMU with PC-side software translating angular motion into smooth mouse movement.
- Implemented hardware-triggered solenoid recoil with MOSFET protection; housed in a fully 3D-printed chassis integrating trigger and recoil mechanics.

AI-Driven Adaptive Lighting Optimization | ***Python, PyTorch, Scikit-learn*** Spring 2025

- Built an ML pipeline to predict optimal automotive lighting parameters (intensity, beam angle) from environmental inputs using synthetic simulation data.
- Developed a Pandas/Matplotlib dashboard to diagnose dataset quality issues and evaluate model generalization across multiple lighting environments.

Real-Time Object Tracking with YOLO on Raspberry Pi 5 | ***Computer Vision, PyTorch*** Spring 2025

- Deployed a YOLOv8 real-time detection and tracking system on Raspberry Pi 5 with servo-motor camera control, achieving embedded-grade inference speeds on ARM.

Quantum Random Walk for Procedural Generation ***[Link]*** | ***Python, PennyLane*** Fall 2024

- Implemented 2-qubit quantum walk circuits (Hadamard superposition, 200-step simulation on a 10×10 grid) for procedural terrain generation, demonstrating quantum principles in computational graphics.

Breast Cancer Detection with SVM & Decision Trees | ***Scikit-learn, Matplotlib, NumPy*** Spring 2024

- Achieved 96% accuracy on breast cancer classification using SVM and Decision Tree models with ensemble methods, feature engineering, and cross-validation for robust generalization.

Skills

Languages: Python, C/C++, Java, SQL (PostgreSQL), NoSQL (MongoDB), JavaScript, HTML/CSS, LaTeX
ML / AI: PyTorch, Scikit-learn, PennyLane, Qiskit, LangChain, LangGraph, OpenCV
Frameworks & Tools: Django, Flask, Bootstrap, Git, Google Cloud Platform, MATLAB, VS Code
Hardware: ESP32-S3, Arduino, MPU6050, PCB Design (KiCad), 3D Printing (FDM)
Mathematics: Calculus, Discrete Math, Linear Algebra, Numerical Methods, Coding Theory
Algorithms: Greedy, Divide & Conquer, Dynamic Programming, Graph Algorithms

Extracurricular Courses

Harvard CS50 ***[Certificate]*** Harvard / edX
Harvard CS50 AI with Python ***[Certificate]*** Harvard / edX
Machine Learning Specialization — Andrew Ng ***[Certificate]*** Coursera / DeepLearning.AI / Stanford
Microsoft Azure AI Fundamentals (AI-900) ***[Certificate]*** Microsoft
Microsoft Azure Fundamentals (AZ-900) ***[Certificate]*** Microsoft